Tuning Diffusion Docking for Structure-Based Drug Design of Allosteric Ligands to the Kinome

NYU

SCCPC

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Objective

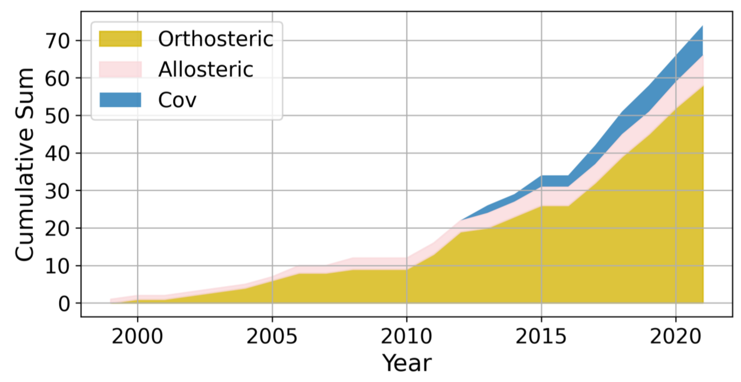
- Develop rational computer-aided drug design (CADD) platform
- Improve blind docking methods for allosteric structure-based drug design (SBDD)

Background

- Use SBDD to improve hit-finding

Case study: Kinases

- Allosteric ligands are promising therapeutics: specificity, non/un-competitive, saturable, cooperative regulation
- Orthosteric inhibitors are most of the FDA-approved kinase small molecule drugs[1]



- Kinase inhibitor types are well-defined

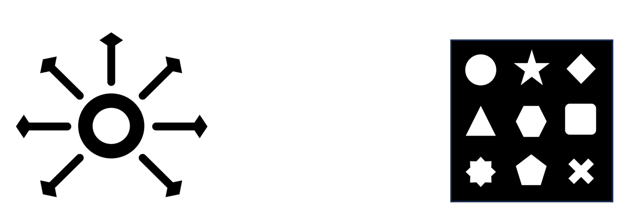
Use case

- An accurate blind docking method can be used to filter away orthosteric docked poses for reduced false positive rate

Current limitations

- The primary failure mode of DiffDock is that it tends to predict orthosteric inhibitors
- There are more orthosteric than allosteric ligands in the training set

Allosteric ligand design essentials

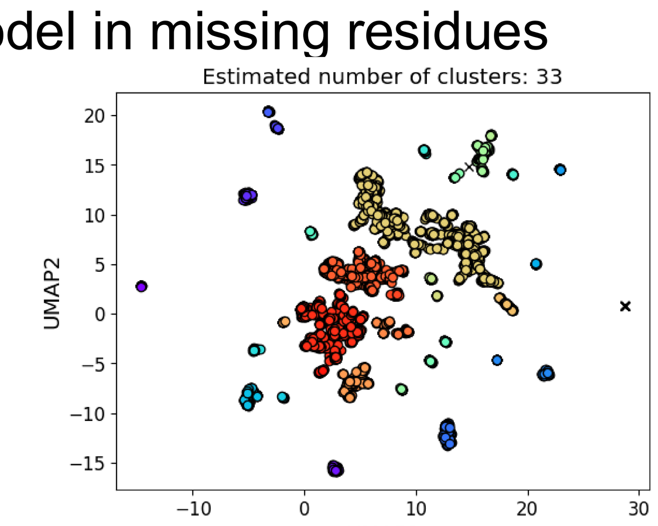


Expand beyond orthosteric site

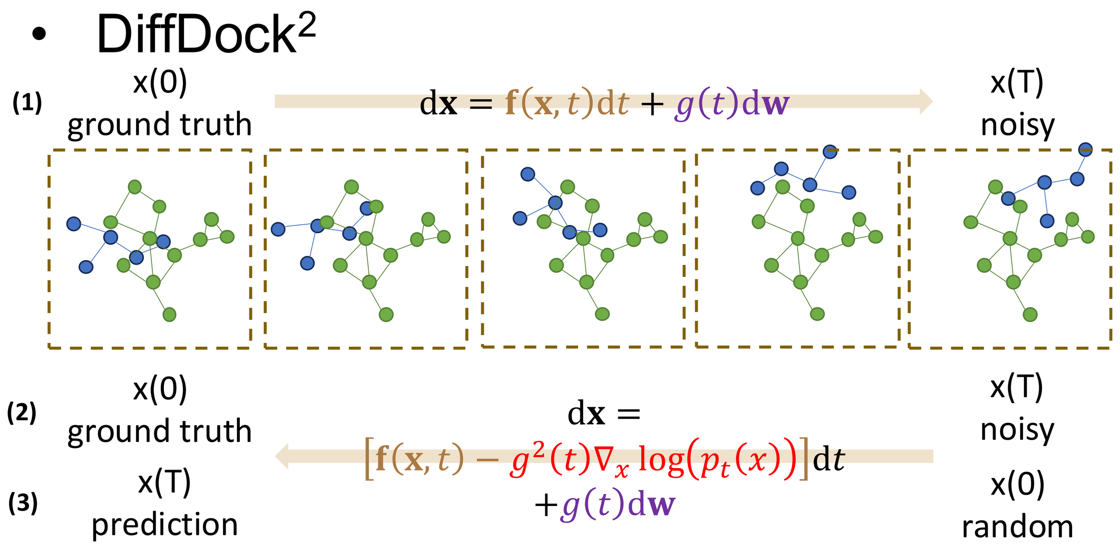
Protein–ligand conformational heterogeneity

Methods

- Preprocessed ligand files
  - Protonated using obabel (pH=7.4)
  - Correct chemistry errors with RDKit including those involving imidazole, tetrazole, amino, and carboxyl groups.
  - Preserve bond types with SDF file
- Protein loop reconstruction
  - Use nearest neighbors as template to model in missing residues

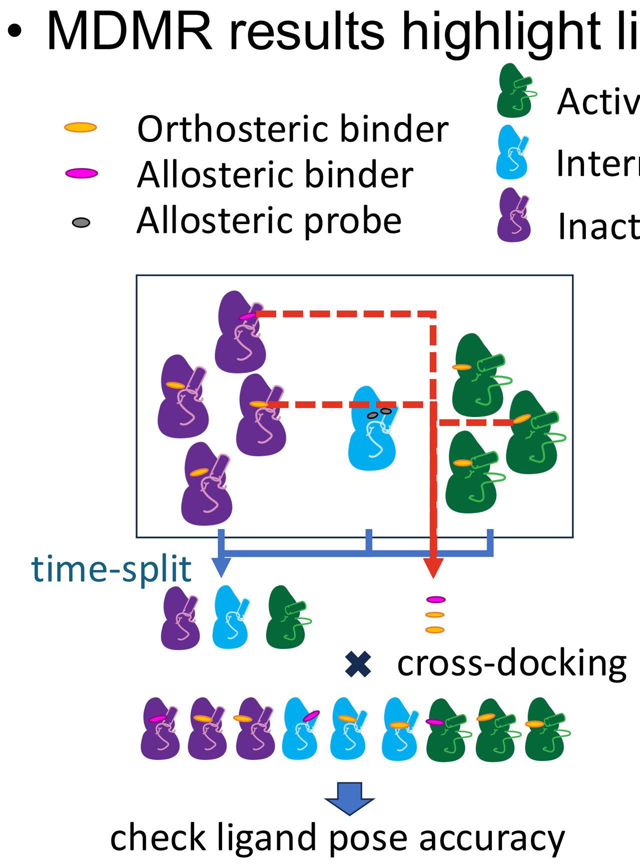


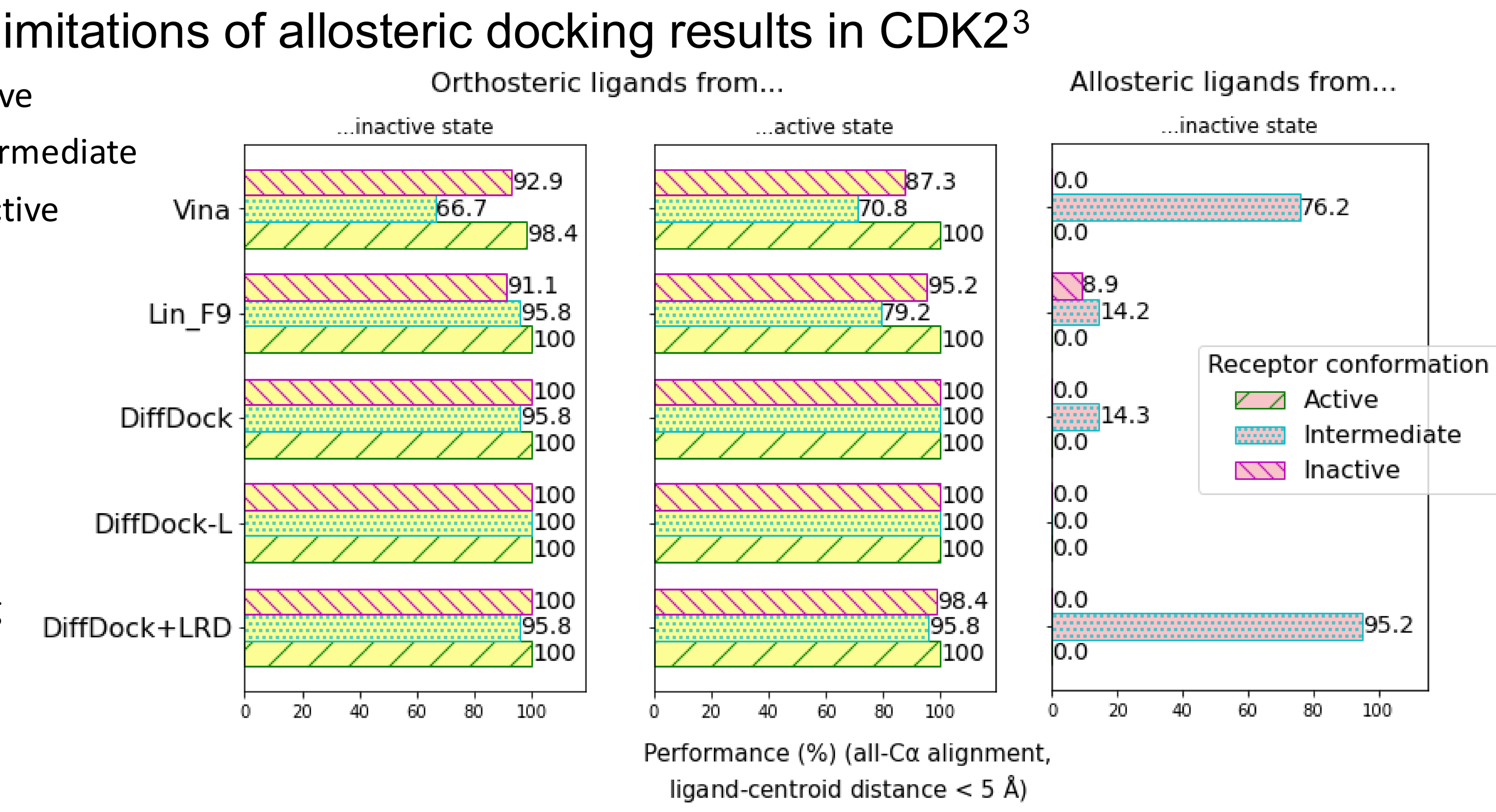
- DiffDock[2]



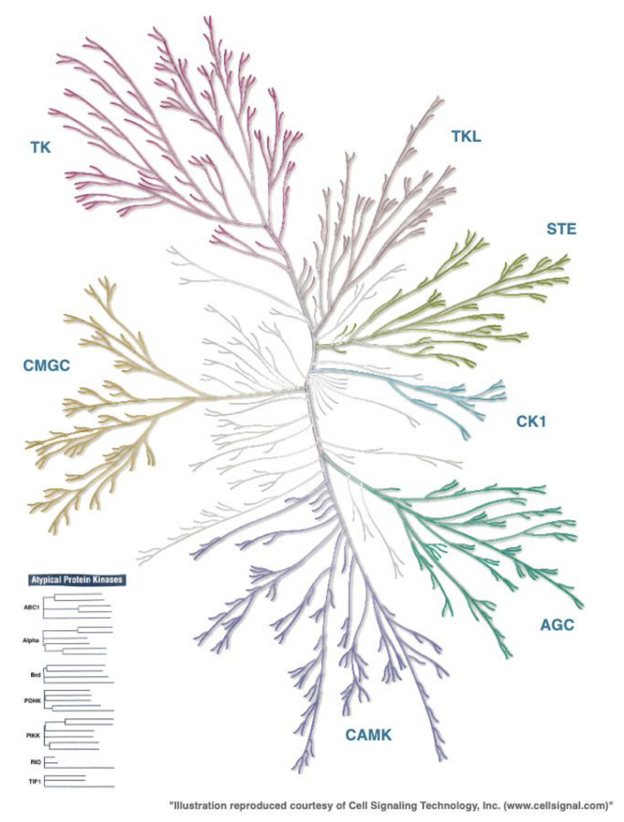
Highlights

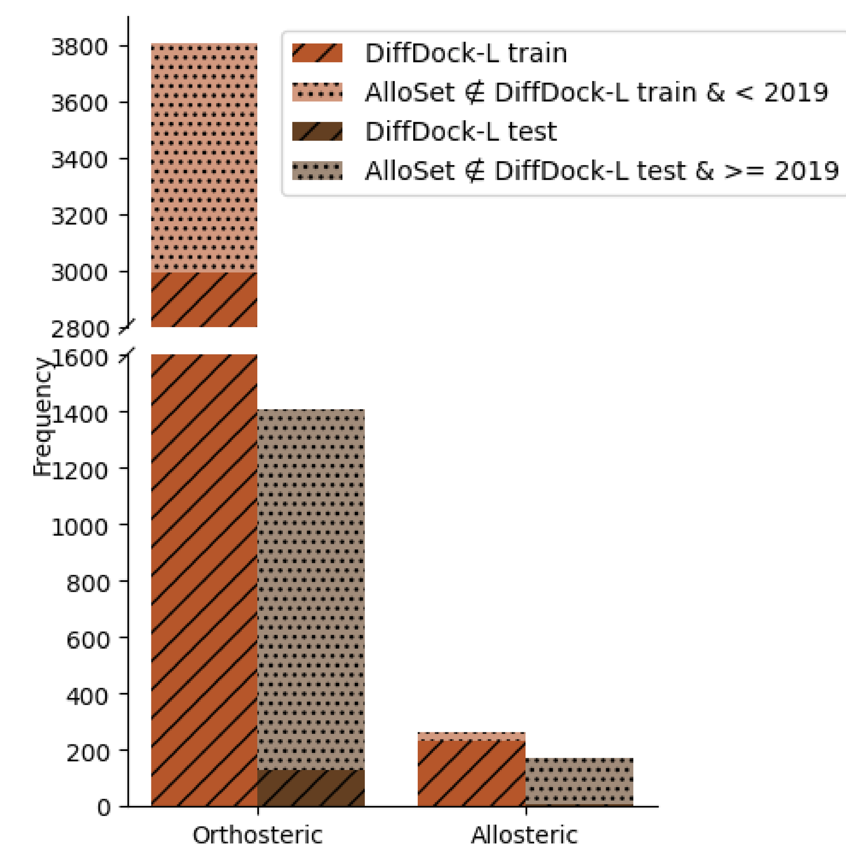
- MDMR results highlight limitations of allosteric docking results in CDK2[3]

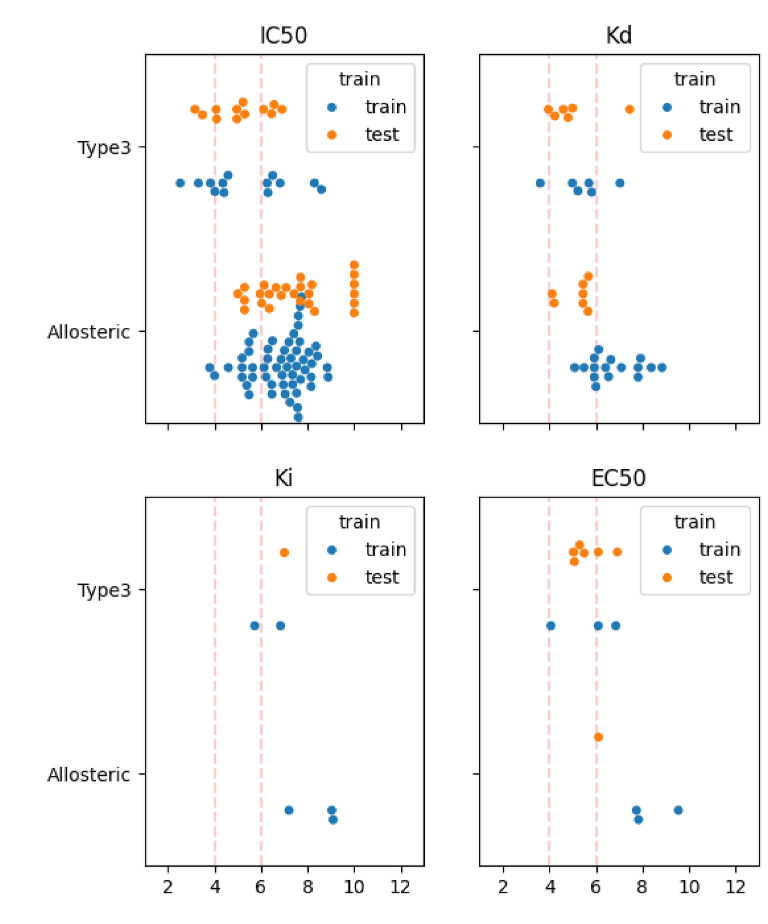




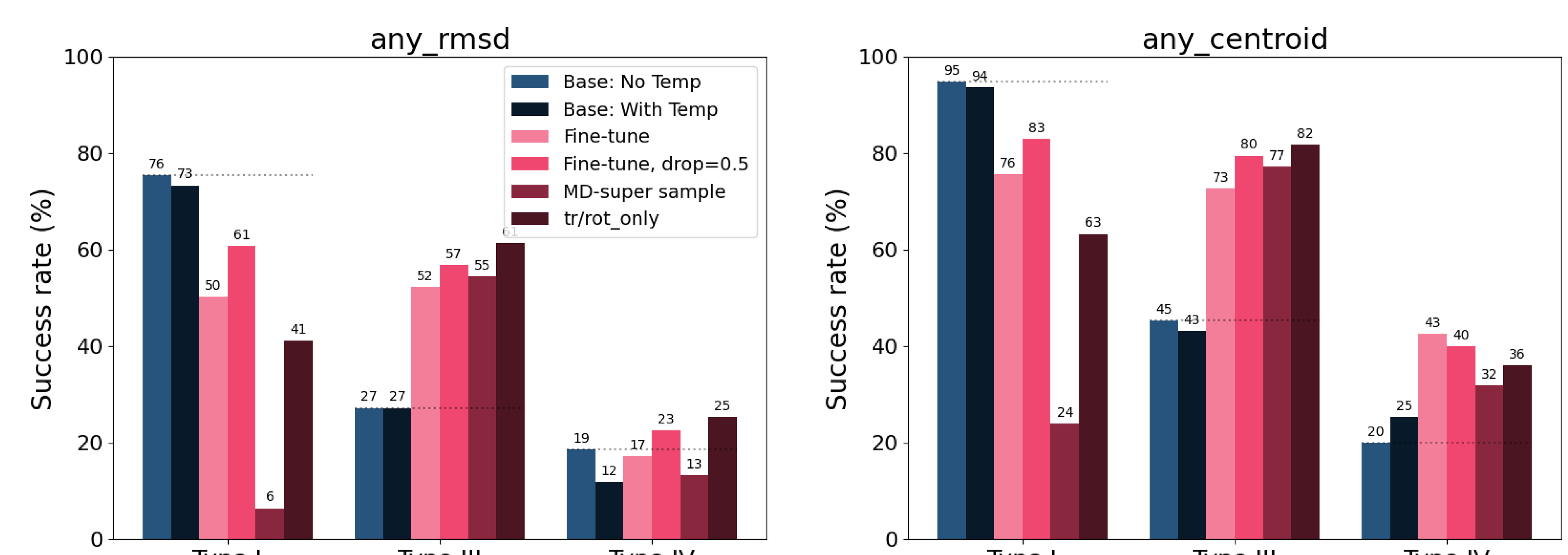
- Kinome-wide dataset generation[4,5]

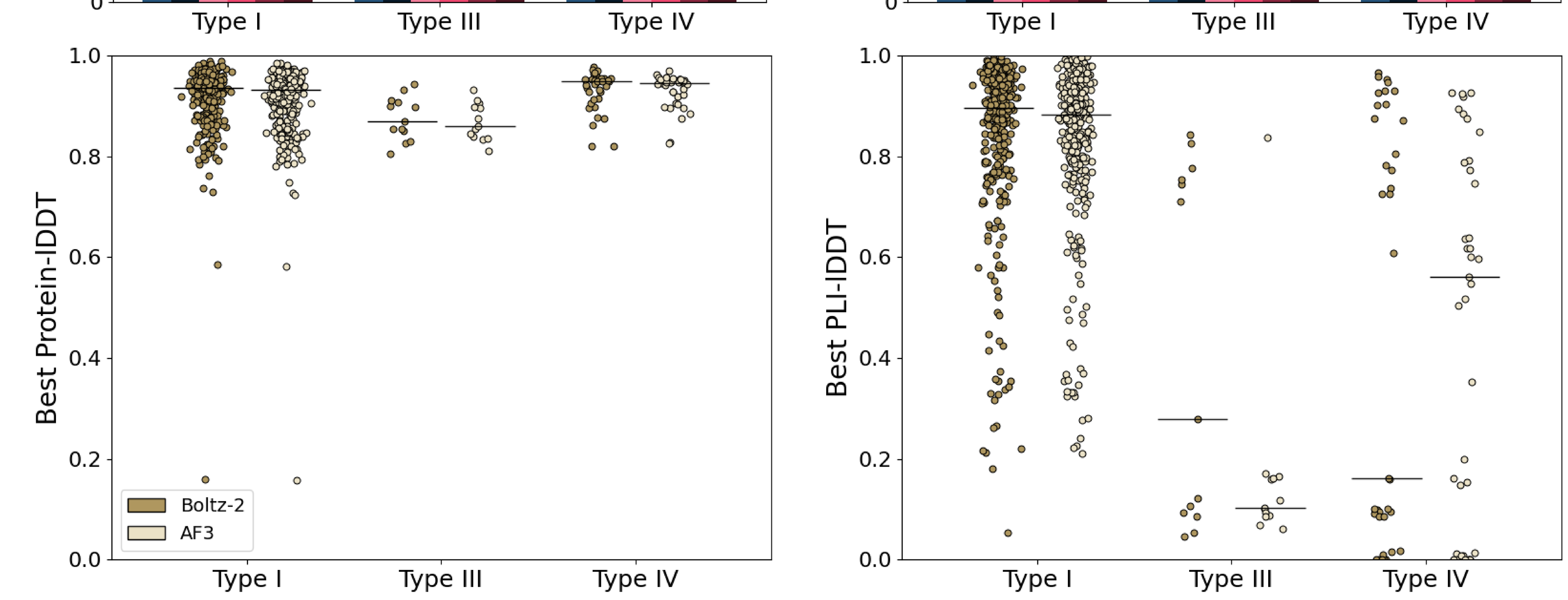






- Results





Future steps

- Improve docking pose accuracy and physical validity
- Enhance conformational diversity of protein structure predictions
- Improve sampling diversity of pose predictions
- Assess within the context of high-throughput screening with diverse library
- Scoring function development

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[1]Pan, Y. & Mader, M. M., *J. Med. Chem.* **65**, 5288–5299 (2022).
[2]Corso, G., Stärk, H., Jing, B., Barzilay, R. & Jaakkola, T. Preprint at http://arxiv.org/abs/2210.01776 (2022).
[3]**Chen, E. A**. & Zhang, Y. *J. Chem. Inf. Model.* **65**, 3737–3748 (2025).
[4]Kanev, G. K., *et al. Nucleic Acids Res.* **49** D562–D569 (2020).
[5]Modi, V. & Dunbrack, R. L., Jr. *Nucleic Acids Res.* **50**, D654–D664 (2022).